Practice Exam - 1

Question 12 of 117

A 72 year-old man presents with NYHA functional class III HFpEF with complaints of shortness of breath and lower extremity edema. Echocardiography shows a LVEF of 50%, left ventricular internal diastolic dimension of 5.0 cm, left ventricular posterior wall diameter in diastole 1.2 cm, bi-atrial enlargement, with apical and septal sparing on global strain imaging. Cardiac magnetic resonance demonstrates diffuse sub-endocardial late gadolinium enhancement in a non-coronary artery territory distribution. Neurologic evaluation was negative. Transthyretin genetic testing does not suggest any pathogenic mutation. Technetium-99m pyrophosphate scintigraphy demonstrates grade 3 myocardial uptake with no evidence of monoclonal protein.

Which of the following disease modifying therapies would be the best therapeutic option to improve this patient’s clinical outcomes?

A. Tafamidis

B. Patisiran

C. Diflunisal

D. Inotersen

Commentary References

Testing Point: The Board-certified AHFTC specialist should select appropriate disease-modifying therapy for transthyretin amyloid cardiomyopathy.

Correct Answer: A. Tafamidis

Rationale: Based on scans, presenting symptoms and genetic testing, this patient carries a diagnosis of wild-type transthyretin (ATTRwt) cardiomyopathy with primary symptoms of heart failure without polyneuropathy. The use of Food and Drug Administration–approved disease-modifying therapy is based on the presence of cardiomyopathy and polyneuropathy and the distinction between hereditary amyloid transthyretin (ATTRv) and ATTRwt amyloidosis. In patients with predominantly cardiac disease resulting from ATTRv or ATTRwt, tafamidis is indicated in those with NYHA class I to III symptoms, and early initiation appears to slow disease progression, making Answer A (Tafamidis) the best answer.

Patients with ATTRv and polyneuropathy should be considered for TTR silencing therapy with patisiran (Answer B) or inotersen (Answer D); currently, neither is indicated for ATTRv-cardiomyopathy (CM) without polyneuropathy or in ATTRwt-CM.

In patients with ATTRv-CM with polyneuropathy, the choice between therapeutic agents is based on accessibility and side-effect profile, thus making Answers B and D incorrect.

Diflusinal (Answer C) may be considered with caution for off-label therapy for asymptomatic ATTR carriers, for patients with ATTR-CM who are not eligible for TTR silencers, or for patients with ATTR-CM who are intolerant of or cannot afford tafamidis, thus making Answer C a less suitable answer than answer A.